PROJECT PROPOSAL

The Impact of Pricing and Ratings on Video Game Popularity

Video games have become one of the largest entertainment industries in the world, and understanding what makes a game popular is more relevant than ever. This project focuses on two factors that directly shape player decisions: price and ratings. The central question is: *How do pricing and ratings influence the popularity of video games?*

The primary dataset is the Steam Games Dataset from Kaggle which includes over 120,000 games with variables such as title, genre, price, release date, developer, and user review counts. Since Steam does not share sales data publicly, review counts will serve as a proxy for popularity, as they reflect genuine player engagement.

https://www.kaggle.com/datasets/fronkongames/steam-games-dataset

The second dataset, Video Game Sales, contains around 16,598 records with platform, genre, publisher, and global sales figures. This allows for an additional perspective by comparing review-based popularity with actual market performance.

https://www.kaggle.com/datasets/gregorut/videogamesales

A third dataset from the Steam Spy API may be used as a supplementary source if needed, providing ownership estimates and playtime data as further indicators of engagement.

https://www.kaggle.com/datasets/muhammadaqeelkabir/steam-games-dataset-steamspy-api

The datasets will be merged based on game titles. Some mismatches are expected due to naming differences, but the overlapping records should be sufficient for meaningful analysis. Data preprocessing will be handled in Python and will include cleaning missing values, standardizing titles, and merging the datasets into a unified structure.

Popularity is defined as the number of user reviews on Steam. Price is taken directly from the listed game price, and ratings are measured using the Critic Score and User Score variables from the Video Game Sales dataset. Together, these variables will allow a data-driven investigation of how pricing and ratings relate to video game popularity and overall market performance.